Paper 1 (SDG 4.1.4) - Complete Model Equations and Data Map

A Bayesian cohort model for out-of-school rates | Paper notation mapped to the current Stan/NumPyro implementation | 28 September 2026

ADMIN-ONLY FIT: RUNNABLE | FULL SURVEY FIT: DATA INCOMPLETE | FORMULAS: DOCUMENTED | FINAL REPRODUCTION: PENDING

1. Model objective and observation types

The latent quantity is the out-of-school rate for country *c*, relative age *g*, and entry/cohort year *e*. The model combines administrative observations and household/census survey observations while allowing administrative error, regional error, survey bias and survey non-sampling error.

$$\kappa_{c,g,e} = \Phi^{-1}(p_{c,g,e})$$

Latent out-of-school rate on the probit scale; $p_{c,g,e} = \Phi(\kappa_{c,g,e})$.

Observation	Paper/code symbol	Meaning	Data source
Administrative rate	A_i	Raw enrolment/population-derived rate; may be negative before the paper's exclusion cutoff.	UIS internal enrolment + WPP/Eurostat/SingStat population.
Survey rate	K_j	Survey/census estimate transformed to probit space, with effective sample size.	DHS, MICS, IPUMS, WIDE and country-specific survey sources.

2. Cohort structure: latent mean and cohort change

The entry-age baseline process is represented by a spline over entry/cohort year. The current implementation uses the basis matrices B1 and B5 constructed by `spline_basis.py`.

$$M_{c,e} = \Sigma_{k=1}^{N_{K1}} \mu^*_{c,k} B1_{e,k}$$

$$\mu^*_{c,k} \sim N(\gamma_c + \mu^*_{c,k-1}, \sigma_\mu^2), \quad k > 1$$

$$\gamma_c \sim N_-(0, \sigma_{\gamma,r(c)}^2)$$

Country drift is constrained to be non-positive; r(c) is the UIS SDG region assigned to country c.

$$\mu^*_{c,1} \sim N(0, 2^2)$$

The paper anchors the baseline at calendar year 2000; the implementation uses the first knot of the backward-extended entry-year grid and documents this as a reproduction convention.

3. Age-by-cohort change: two-dimensional P-spline

For ages after the entry age, the change from the baseline is represented by a cumulative age-by-entry-year surface. The free coefficient surface is $\alpha_{c,g,k}$.

$$\Theta_{c,g,e} = \Sigma_{k=1}^{N_{K5}} B5_{e,k} \Sigma_{h=1}^{g} \alpha_{c,h,k}$$

$$\kappa_{c,0,e} = M_{c,e}; \quad \kappa_{c,g,e} = M_{c,e} + \Theta_{c,g,e}, \quad g > 0$$

$$\alpha_{c,g,k} - \alpha_{c,g-1,k} \sim N(0, \tau_{\alpha 1}^{-1})$$

$$\alpha_{c,g,k} - \alpha_{c,g,k-1} \sim N(0, \tau_{\alpha 2}^{-1})$$

$$\Sigma_g \alpha_{c,g,k} \sim N(0, 2^2)$$

These three equations are implemented as NumPyro factor penalties corresponding to the Stan target increments.

4. Administrative likelihood (Admin-only and Full models)

For an administrative observation i, the model is on the original rate scale. The regional error term is shared by observations from the same UIS SDG region.

$$A_i \sim N(\Phi(\kappa_{c(i),g(i),e(i)}), \sigma_{adm}^2 + \omega_{r(i)}^2)$$

$$A_i = 1 - E_{c,y,a,s} / P_{c,y,a,s}$$

E is total enrolment summed across the applicable education levels; P is the matching single-age population denominator.

$$\sigma_{adm} \sim \text{HalfNormal}(\sigma_\sigma); \quad \omega_r \sim \text{HalfNormal}(\sigma_\sigma)$$

Code field	Formula role	Input/derived variable
`A_obs`	A_i	`oosc_rate` from processed administrative input.
`A_country`	c(i)	Country index from UIS country code.
`A_region`	r(i)	UIS `SDG_*` region index.
`A_age`	g(i)	Relative age index, not raw calendar age.

`A_ey_idx`	e(i)	Entry/cohort year grid index from year and age.

5. Survey likelihood (Full model only)

Survey observations are converted to the probit scale. Survey series have country-specific bias, while survey waves have non-sampling error.

$$K_j = \Phi^{-1}(q_j)$$

$$K_j \sim N(\kappa_{c(j),g(j),e(j)} + \beta_{c(j),s(j)},\ (\nu_j^2 + \eta_{w(j)}^2) / f_j^2)$$

$$\beta_{c,s} \sim N(0, \sigma_{\beta,s}^2);\ \ \eta_w \sim \text{HalfNormal}(\sigma_\sigma)$$

q is the observed survey rate; f is the probit density at K; nu is the survey standard-error input on the relevant scale.

Code field	Formula role	Required survey content
`K_obs`	K_j	Survey OOSC/attendance estimate transformed using the normal inverse CDF.
`f_obs`	f_j	Normal density evaluated at K_j.
`nu`	ν_j	Survey standard-error input.
`S_series`, `S_wave`	s(j), w(j)	Survey family/series and survey wave identifiers.

6. Shared priors and variance hierarchy

$$\sigma_{\gamma,r} \sim \text{HalfNormal}(0.02);\ \ \sigma_\mu \sim \text{HalfNormal}(0.1);\ \ \tau_{\alpha 1}, \tau_{\alpha 2} \sim \text{HalfNormal}(1)$$

$$\sigma_\sigma \sim \text{HalfNormal}(0.01);\ \ \sigma_\beta \sim \text{HalfNormal}(0.1)$$

The model is fitted separately by sex population in the paper's specification. The current data pipeline must therefore use consistent sex coding and decide whether the MF/total series is a separate fit or an aggregate fit.

7. Data-to-variable map

Paper/model quantity	Administrative dataset	Survey dataset	Processing requirement
Country, year, age, sex	UIS enrolment; WPP/Eurostat/SingStat population.	DHS PR, MICS member/household files, IPUMS person records, WIDE aggregates.	Common country, year, age and sex schema.

Enrolment E	`uis_internal_enrolment_single_age_2000_2022.csv`.	Not applicable.	Select correct EMC indicators; sum across education levels by cell.
Population P	WPP, Eurostat, SingStat.	Survey denominator/weight where applicable.	Match country-year-age-sex and convert units to persons.
School attendance q	Not applicable.	DHS current attendance; MICS education module; IPUMS SCHOOL; WIDE education indicators; Colombia ECV P8586.	Harmonise attendance definition and retain survey sample size/uncertainty.
Region r	UIS country-group reference/membership, selecting `SDG_*`.	Same country mapping.	One region label and integer region_id for every country.

8. Admin-only versus full model implementation

Feature	Admin-only preliminary fit	Full paper fit
Administrative likelihood	Included.	Included.
Survey likelihood	Empty by design.	Included after survey harmonisation.
Survey bias beta	Parameters remain structurally declared but have no survey observations to identify them.	Estimated from survey observations and priors.
Interpretation	Pipeline, convergence and administrative cohort-pattern test.	Reproduction candidate after all survey sources and paper vintages are included.

> **Do not over-interpret the admin-only fit:** it cannot estimate the information contributed by household surveys, survey bias or survey non-sampling error. It is nevertheless valuable for debugging and for an administrative-only sensitivity analysis.

9. Current data gaps and execution order

1. Create the UIS SDG region mapping and add `region`/`region_id` to processed administrative data.
2. Run `Admin Only/admin_only_model.py` with small warmup and sample counts.
3. Validate MCMC diagnostics, coverage and negative-rate handling.
4. Harmonise MICS and available WIDE indicators.
5. Complete Colombia ECV by adding the person/demographic file to the education module.
6. Obtain Papua New Guinea HIES person-level data.
7. Obtain DHS PR files and IPUMS extracts with codebooks.
8. Construct the full survey arrays K_obs, f_obs, nu, S_country, S_age, S_ey_idx, S_series and S_wave.
9. Fit the full model and compare posterior outputs with ROFST.MOD validation estimates.

> **Code locations:**
> `D. Code/D.1. Paper 1 - OOSC Cohort Model (4.1.4)/Admin Only/admin_only_model.py`
> `D. Code/D.1. Paper 1 - OOSC Cohort Model (4.1.4)/Python/oosc_model_numpyro.py`